



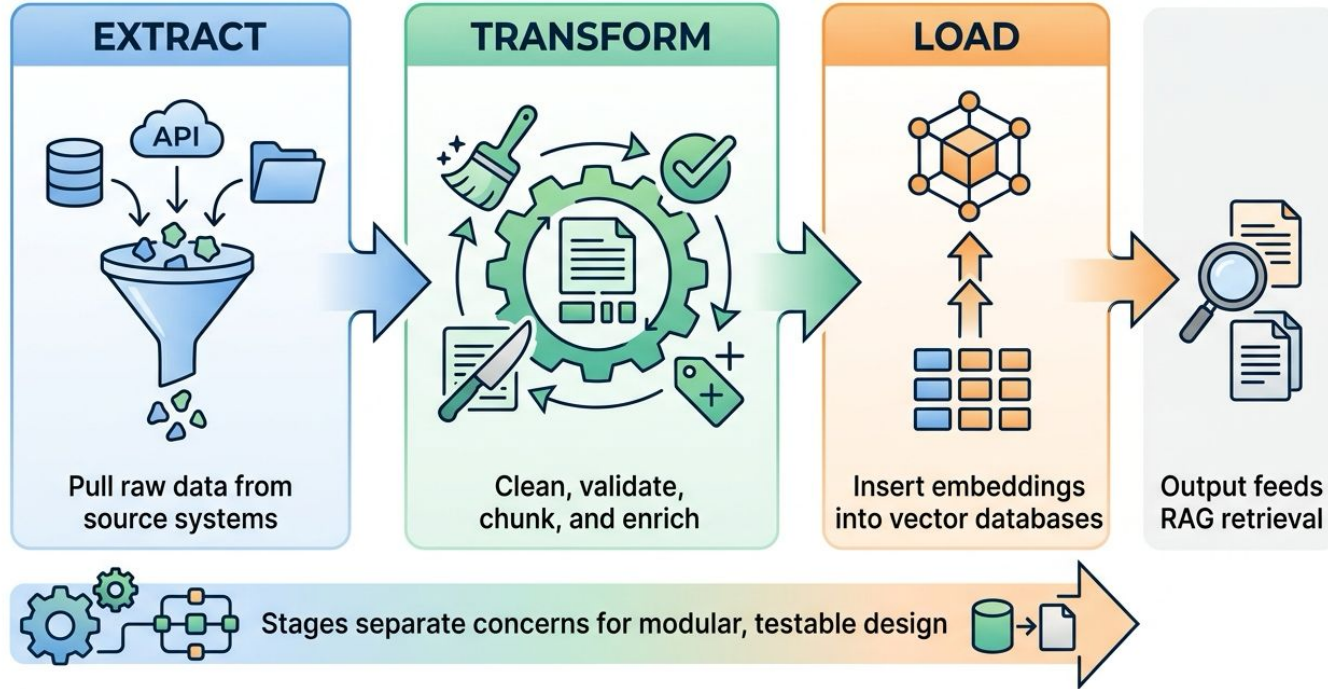
U.S. General Services
Administration



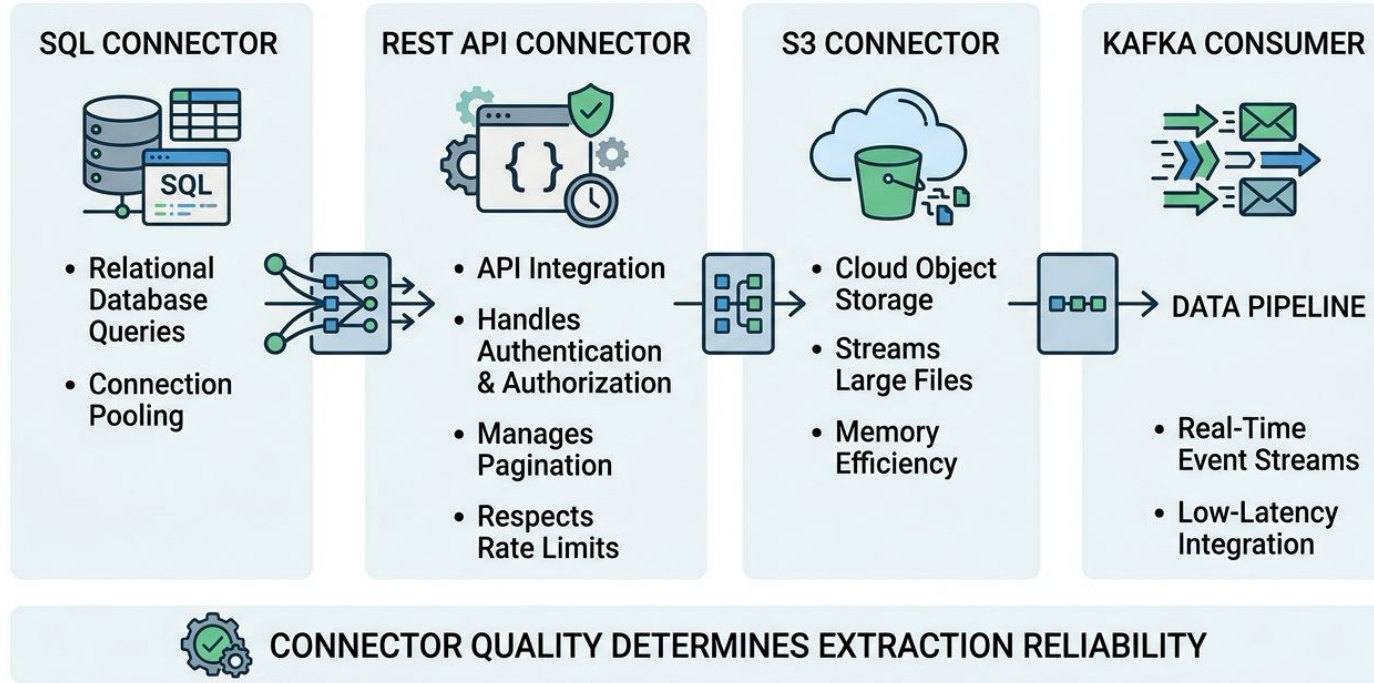
Chapter 6.3A: ETL Pipeline Fundamentals

US General Service Administration
AI Community of Practice - March 2026

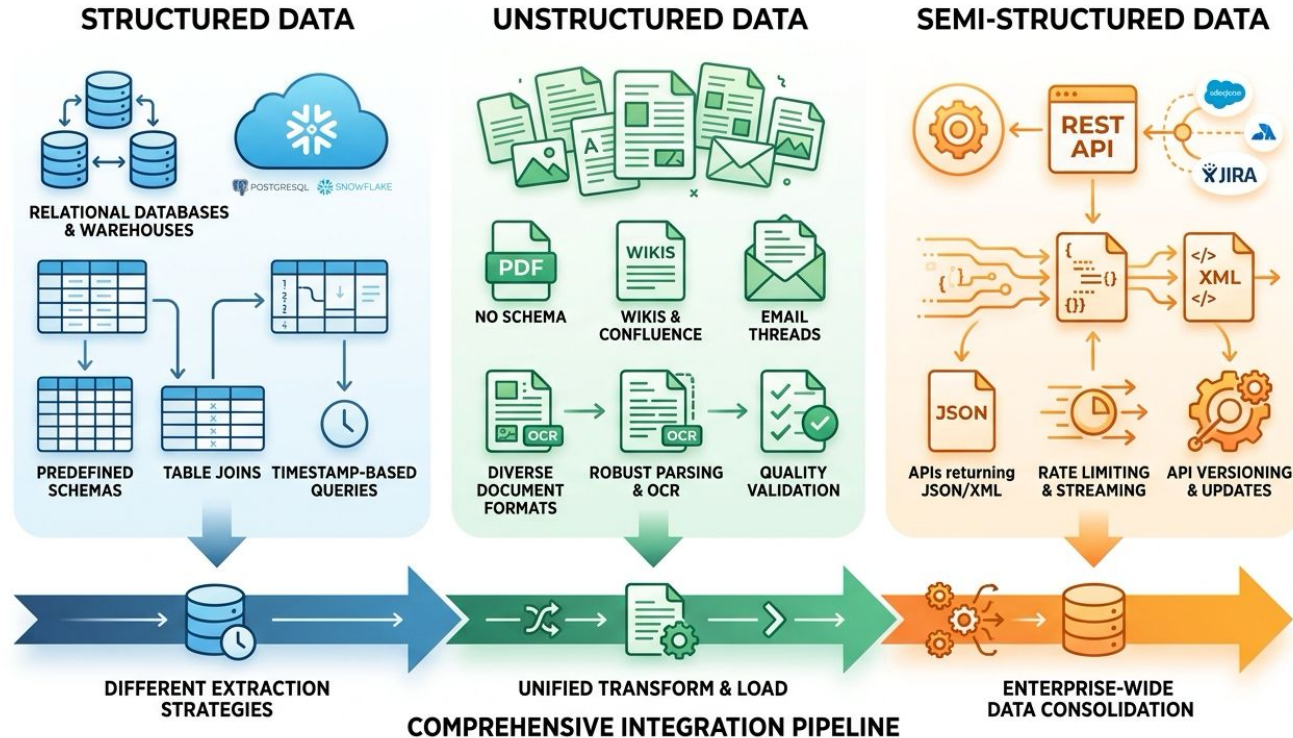
The ETL Architecture -- Three Stages, One Goal



Extract Phase -- Connectors Bridge the Source Gap



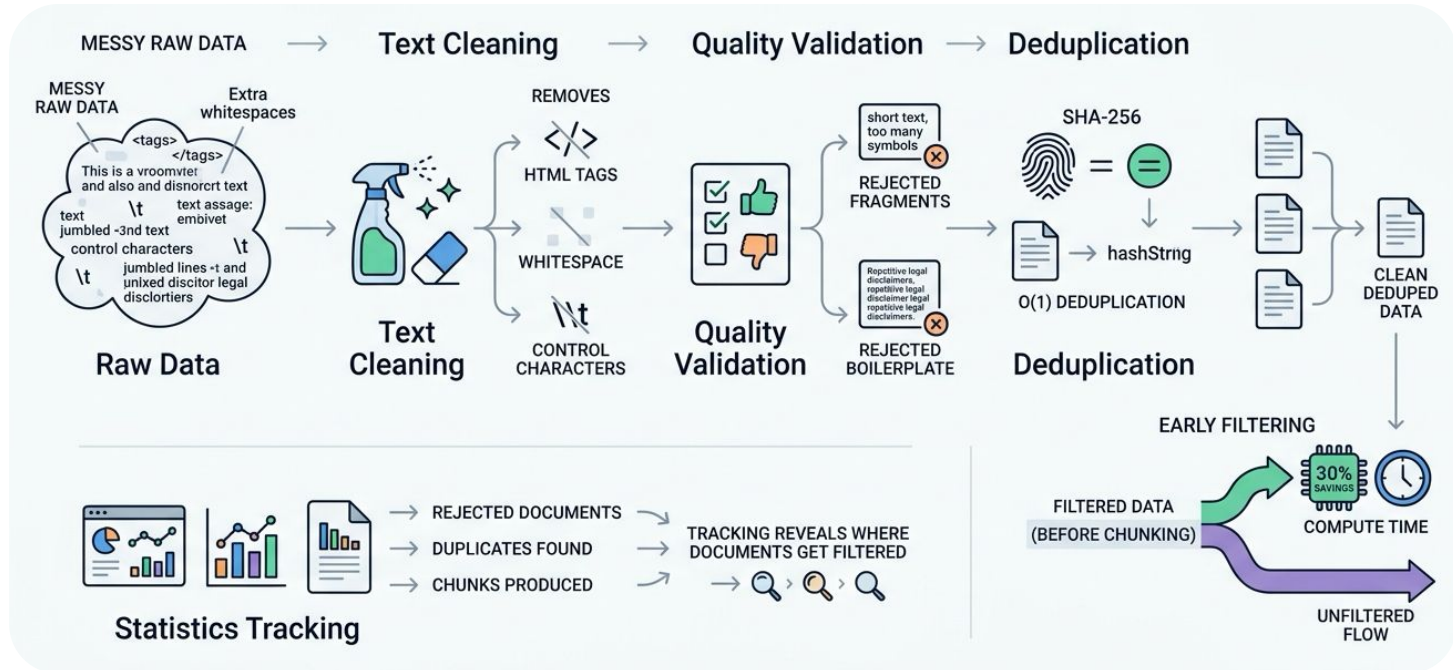
Data Source Categories



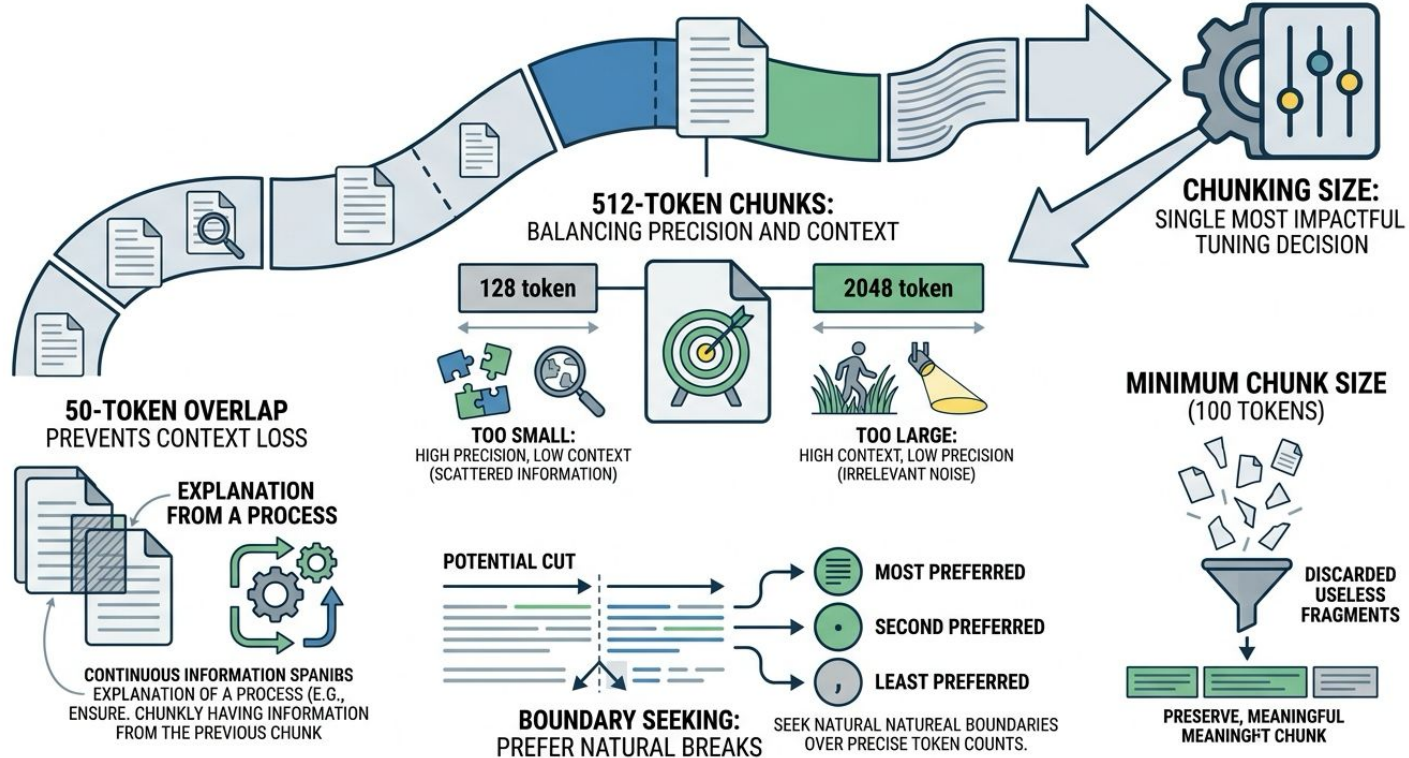
Incremental Updates -- Process Only What Changed

- Full reprocessing becomes infeasible at enterprise scale
 - Timestamp-based filtering detects new or modified records
 - Reduces daily ETL runtime from hours to minutes
 - File systems use modification time for change detection
 - Trade-off: added complexity in change-tracking logic
- 

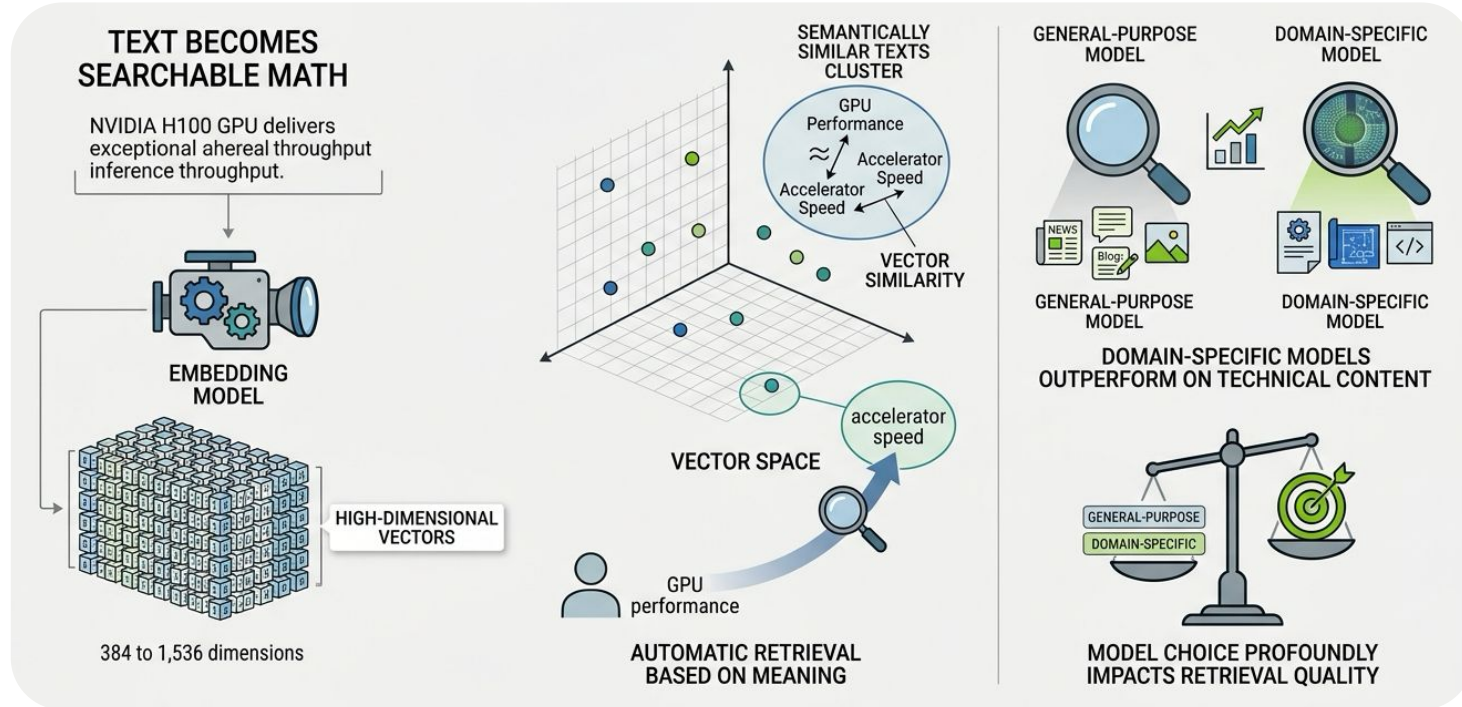
Transform Phase



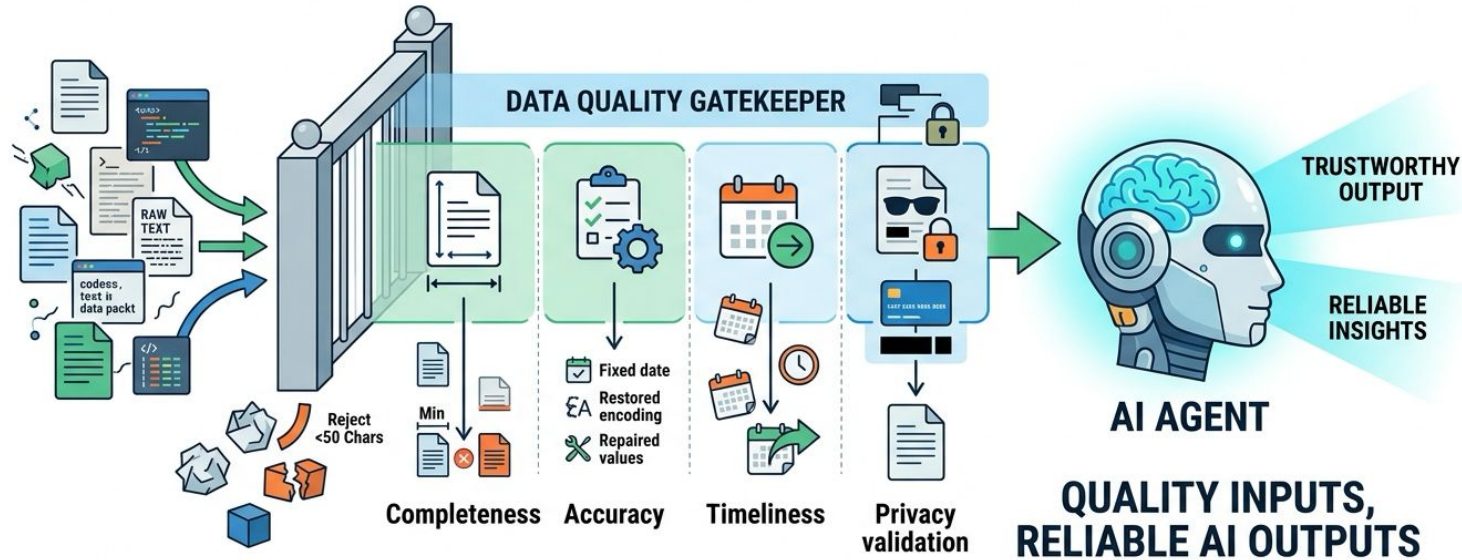
Chunking Strategy



Vector Embeddings



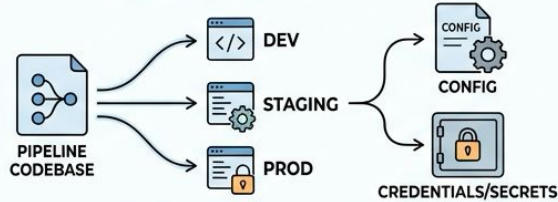
Data Quality -- The Gatekeeper Your Agent Needs



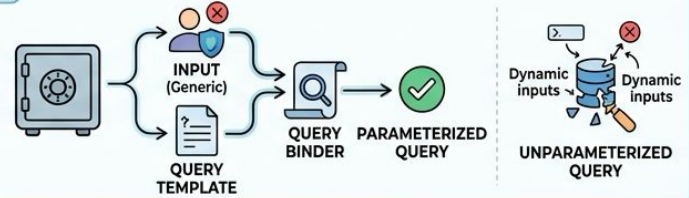
Pipeline Orchestration and Error Handling



CONFIGURATION SEPARATION



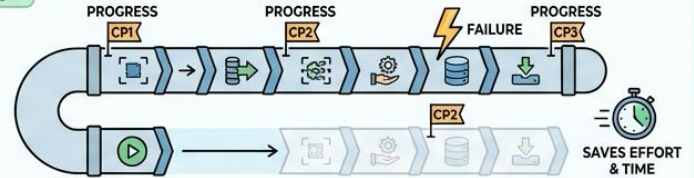
PARAMETERIZED QUERIES (SQL INJECTION PREVENTION)



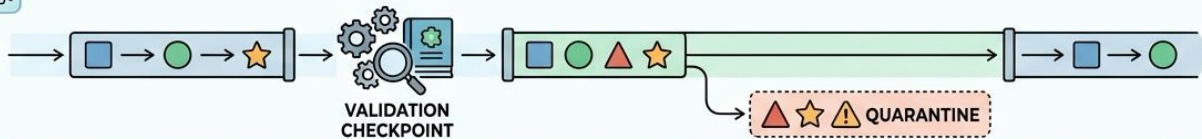
RETRY LOGIC: TRANSIENT vs. PERMANENT FAILURES



CHECKPOINTING & PROGRESS PERSISTENCE

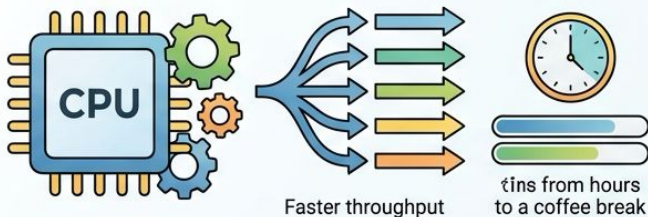


SCHEMA VALIDATION: EARLY CATCH



Production Enhancements -- Scaling the Pipeline

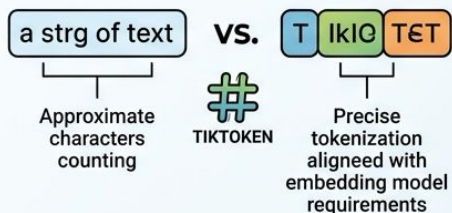
PARALLEL PROCESSING



FUZZY DEDUPLICATION



TOKEN-ACCURATE COUNTING



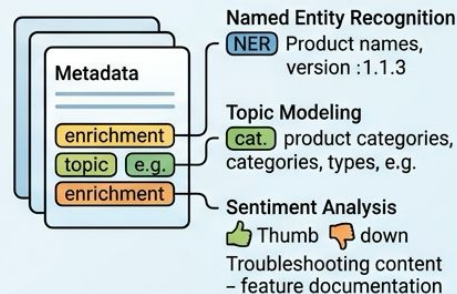
Ensures accuracy, especially for code quote for code or non-English text

SEMANTIC-AWARE SPLITTING



More coherent and meaningful document chunks

METADATA ENRICHMENT



Key Takeaways

- ETL is the backbone connecting enterprise data to agent knowledge
 - Three stages: Extract sources, Transform quality, Load vectors
 - Incremental updates and quality gates are non-negotiable at scale
 - Chunking strategy is the highest-impact tuning decision
 - Next: Chapter 6.3B covers the Load phase and integration
- 